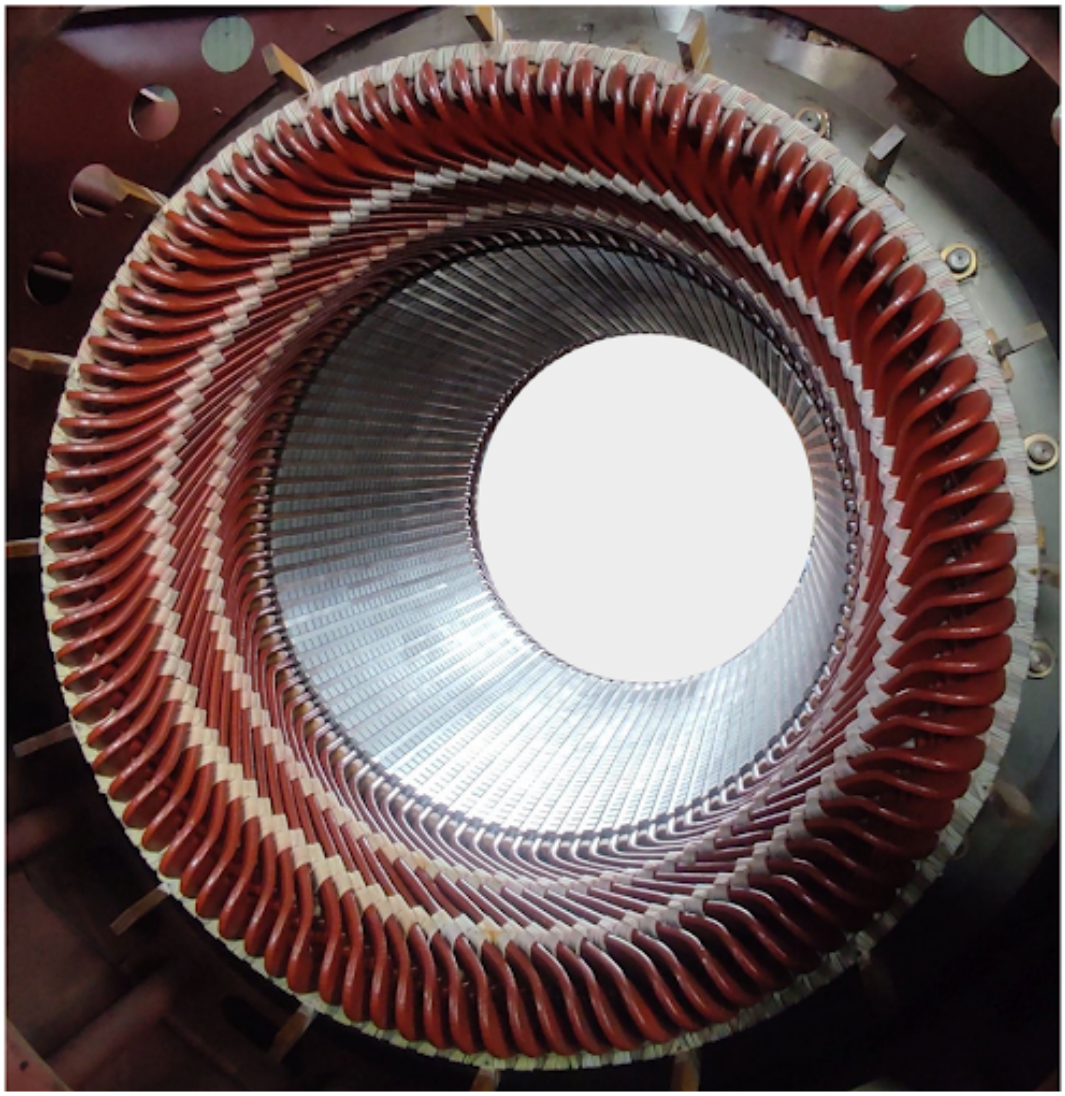
TechGM
Condition Monitoring Solutions

Report - Backup Generator

Report Details	
Customer	Acme Industries
Tag ID	MTR-005
Make	Siemens
Rating	45.5 kW | 400 V | 1450 RPM
Report Date	Aug 21, 2026



** Cover page photo is for illustration purpose and does not represent the test object in this report.*

Date of Testing	Aug 21, 2026
Report Date	Aug 21, 2026
Prepared By	Super Admin
Order No	MTR-005
Order Date	N/A

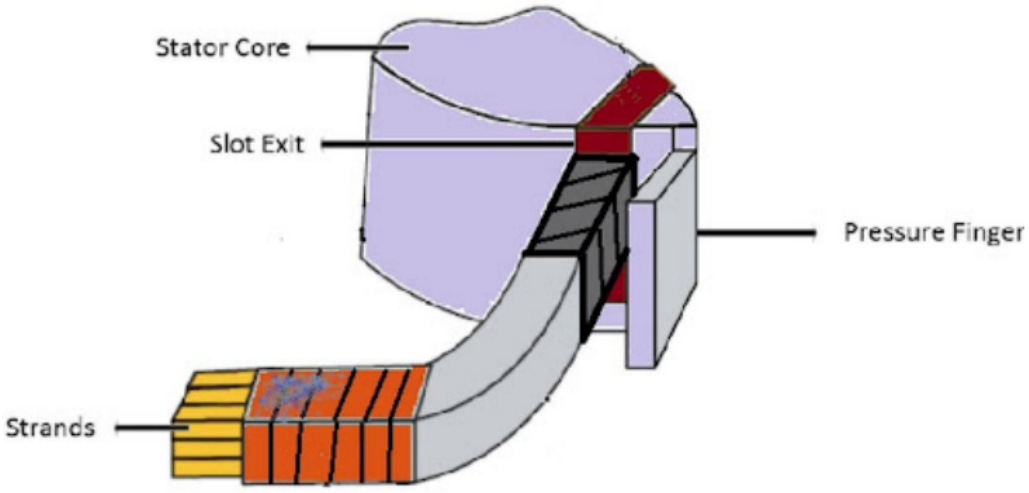
Color Legend

- **Health:** Low likelihood of presence of severity of defects and abnormalities. Follow normal maintenance program.
- **Tolerable:** Considered acceptable for long-term continuous operation. Follow normal maintenance program and review recommendations.
- **Weak:** Not acceptable for long-term continuous operation. Plan for overhaul and maintenance.
- **Critical:** Machine to be put into service only after immediate maintenance.

Machine Rating

Serial No	MTR-005
Make	Siemens
Type	1LE1
Year of Mfg	N/A
Rating (kW)	45.5 kW
Power Factor	N/A
Stator Volts (V)	400 V
Stator Current (A)	80 A
Rotor Volts (V)	N/A
Rotor Current (A)	N/A
Class of Insulation	N/A
Connection	N/A
Rated Frequency (Hz)	N/A
Rated Speed (RPM)	1450 RPM

Summary



Contamination	Ground Wall Insulation	Stress Grade System	Corona Protection

TEAM Stress:

Thermal	
Electrical	
Ambient	
Mechanical	

Quick Notes:

- There are contaminants like dry dust and oil/grease along stator windings.
- There is some delamination between tape layers that is initiated resulting in air spaces formation within ground-wall insulation that may accelerate insulation aging.

Next Maintenance Action:

- Overhangs should be cleaned at the earliest.
- Bearings should be inspected for oil/grease spillage on the stator windings.

PDCA Test Results - Raw Data

Time	U Phase		V Phase		W Phase		UVW Phase	
(s)	IR (m©)	Idch (mV)	IR (m©)	Idch (mV)	IR (m©)	Idch (mV)	IR (m©)	Idch (mV)
5	120.00	17600.00	112.00	16820.00	112.00	19800.00	60.15	42000.00
10	245.00	13080.00	238.00	12380.00	246.00	14820.00	113.20	30900.00
15	349.00	9810.00	355.00	8780.00	321.00	10180.00	171.90	18620.00
20	475.00	6950.00	490.00	6720.00	429.00	7950.00	238.00	13950.00
25	596.00	5590.00	728.00	5470.00	540.00	6420.00	298.00	11190.00
30	727.00	4710.00	812.00	4630.00	646.00	5430.00	372.00	8840.00
40	951.00	3643.00	989.00	3582.00	838.00	4126.00	539.00	6220.00
50	1143.00	3032.00	1117.00	2957.00	1019.00	3349.00	694.00	4710.00
60	1321.00	2573.00	1418.00	2599.00	1154.00	2858.00	857.00	3902.00
70	1498.00	2243.00	1582.00	2220.00	1304.00	2467.00	983.00	3280.00
80	1639.00	1972.00	1744.00	1978.00	1439.00	2188.00	1080.00	2848.00
90	1770.00	1796.00	1929.00	1786.00	1564.00	1963.00	1236.00	2532.00
100	1928.00	1612.00	2060.00	1613.00	1689.00	1770.00	1335.00	2280.00
110	2050.00	1509.00	2230.00	1480.00	1803.00	1624.00	1415.00	2089.00
120	2180.00	1390.00	2360.00	1393.00	1924.00	1493.00	1495.00	1880.00
150	2510.00	1139.00	2740.00	1198.00	2270.00	1195.00	1793.00	1557.00
180	2860.00	957.00	3120.00	960.00	2510.00	988.00	2040.00	1314.00
210	3180.00	825.00	3490.00	825.00	2790.00	828.00	2200.00	1142.00
240	3470.00	723.00	3860.00	720.00	3130.00	706.00	2420.00	1009.00
270	3810.00	640.00	4190.00	638.00	3530.00	609.00	2600.00	902.00
300	4080.00	573.00	4470.00	568.00	3810.00	541.00	2780.00	815.00
330	4470.00	511.80	4820.00	505.50	4010.00	478.00	3000.00	742.00
360	4650.00	464.80	5130.00	459.50	4240.00	429.00	3100.00	681.00
390	5110.00	424.90	5480.00	417.90	4650.00	382.10	3250.00	629.00
420	5330.00	390.30	5740.00	382.70	4890.00	348.20	3330.00	576.40
450	5620.00	360.00	6060.00	351.80	4910.00	317.20	3460.00	546.20
480	5890.00	333.90	6340.00	324.10	5150.00	291.40	3620.00	512.30
510	5840.00	309.60	6540.00	299.20	5330.00	269.00	3670.00	479.50
540	6100.00	288.10	6810.00	280.90	5340.00	249.60	3870.00	453.60
570	6360.00	268.70	7200.00	262.20	5450.00	232.90	3870.00	429.30
600	6640.00	251.20	7350.00	245.70	5520.00	217.90	3980.00	408.20
630	6670.00	235.60	7690.00	230.90	5570.00	205.00	4180.00	389.30
660	6970.00	221.50	8080.00	216.70	5870.00	193.50	4140.00	371.60
690	6890.00	208.60	8270.00	203.70	6000.00	183.50	4290.00	355.20
720	7130.00	196.80	8930.00	192.20	6160.00	174.50	4310.00	340.40
750	7460.00	185.40	8510.00	181.60	6260.00	165.50	4480.00	326.20
780	7690.00	174.80	8740.00	172.20	6450.00	156.50	4590.00	314.30
810	7890.00	165.20	9080.00	163.40	6590.00	148.50	4670.00	302.00
840	8050.00	155.50	9180.00	154.40	6540.00	139.30	4810.00	290.90
870	8290.00	148.80	9470.00	146.50	6550.00	131.40	4870.00	280.90
900	8480.00	142.30	9620.00	138.80	6620.00	125.70	5080.00	271.00
930	8670.00	135.80	9770.00	131.10	6690.00	120.00	5290.00	261.10
960	8860.00	129.30	9920.00	123.40	6760.00	114.30	5500.00	251.20
990	9050.00	122.80	10070.00	115.70	6830.00	108.60	5710.00	241.30
1020	9240.00	116.30	10220.00	108.00	6900.00	102.90	5920.00	231.40

PDCA Test Results - Calculated Data

Time	U Phase		V Phase		W Phase		UVW Phase	
(s)	IR (μA)	Idch (μA)	IR (μA)	Idch (μA)	IR (μA)	Idch (μA)	IR (μA)	Idch (μA)
5	41.666667	17.600000	44.642857	16.820000	44.642857	19.800000	83.125520	42.000000
10	20.408163	13.080000	21.008403	12.380000	20.325203	14.820000	44.169611	30.900000
15	14.326648	9.810000	14.084507	8.780000	15.576324	10.180000	29.086678	18.620000
20	10.526316	6.950000	10.204082	6.720000	11.655012	7.950000	21.008403	13.950000
25	8.389262	5.590000	6.868132	5.470000	9.259259	6.420000	16.778523	11.190000
30	6.877579	4.710000	6.157635	4.630000	7.739938	5.430000	13.440860	8.840000
40	5.257624	3.643000	5.055612	3.582000	5.966587	4.126000	9.276438	6.220000
50	4.374453	3.032000	4.476276	2.957000	4.906771	3.349000	7.204611	4.710000
60	3.785011	2.573000	3.526093	2.599000	4.332756	2.858000	5.834306	3.902000
70	3.337784	2.243000	3.160556	2.220000	3.834356	2.467000	5.086470	3.280000
80	3.050641	1.972000	2.866972	1.978000	3.474635	2.188000	4.629630	2.848000
90	2.824859	1.796000	2.592017	1.786000	3.196931	1.963000	4.045307	2.532000
100	2.593361	1.612000	2.427184	1.613000	2.960332	1.770000	3.745318	2.280000
110	2.439024	1.509000	2.242152	1.480000	2.773156	1.624000	3.533569	2.089000
120	2.293578	1.390000	2.118644	1.393000	2.598753	1.493000	3.344482	1.880000
150	1.992032	1.139000	1.824818	1.198000	2.202643	1.195000	2.788622	1.557000
180	1.748252	0.957000	1.602564	0.960000	1.992032	0.988000	2.450980	1.314000
210	1.572327	0.825000	1.432665	0.825000	1.792115	0.828000	2.272727	1.142000
240	1.440922	0.723000	1.295337	0.720000	1.597444	0.706000	2.066116	1.009000
270	1.312336	0.640000	1.193317	0.638000	1.416431	0.609000	1.923077	0.902000
300	1.225490	0.573000	1.118568	0.568000	1.312336	0.541000	1.798561	0.815000
330	1.118568	0.511800	1.037344	0.505500	1.246883	0.478000	1.666667	0.742000
360	1.075269	0.464800	0.974659	0.459500	1.179245	0.429000	1.612903	0.681000
390	0.978474	0.424900	0.912409	0.417900	1.075269	0.382100	1.538462	0.629000
420	0.938086	0.390300	0.871080	0.382700	1.022495	0.348200	1.501502	0.576400
450	0.889680	0.360000	0.825083	0.351800	1.018330	0.317200	1.445087	0.546200
480	0.848896	0.333900	0.788644	0.324100	0.970874	0.291400	1.381215	0.512300
510	0.856164	0.309600	0.764526	0.299200	0.938086	0.269000	1.362398	0.479500
540	0.819672	0.288100	0.734214	0.280900	0.936330	0.249600	1.291990	0.453600
570	0.786164	0.268700	0.694444	0.262200	0.917431	0.232900	1.291990	0.429300
600	0.753012	0.251200	0.680272	0.245700	0.905797	0.217900	1.256281	0.408200
630	0.749625	0.235600	0.650195	0.230900	0.897666	0.205000	1.196172	0.389300
660	0.717360	0.221500	0.618812	0.216700	0.851789	0.193500	1.207729	0.371600
690	0.725689	0.208600	0.604595	0.203700	0.833333	0.183500	1.165501	0.355200
720	0.701262	0.196800	0.559910	0.192200	0.811688	0.174500	1.160093	0.340400
750	0.670241	0.185400	0.587544	0.181600	0.798722	0.165500	1.116071	0.326200
780	0.650195	0.174800	0.572082	0.172200	0.775194	0.156500	1.089325	0.314300
810	0.633714	0.165200	0.550661	0.163400	0.758725	0.148500	1.070664	0.302000
840	0.621118	0.155500	0.544662	0.154400	0.764526	0.139300	1.039501	0.290900
870	0.603136	0.148800	0.527983	0.146500	0.763359	0.131400	1.026694	0.280900
900	0.589623	0.142300	0.519751	0.138800	0.755287	0.125700	0.984252	0.271000
930	0.576701	0.135800	0.511771	0.131100	0.747384	0.120000	0.945180	0.261100
960	0.564334	0.129300	0.504032	0.123400	0.739645	0.114300	0.909091	0.251200
990	0.552486	0.122800	0.496524	0.115700	0.732064	0.108600	0.875657	0.241300
1020	0.541126	0.116300	0.489237	0.108000	0.724638	0.102900	0.844595	0.231400

PDCA Test Results - Estimated Parameters

Parameter	U Phase	V Phase	W Phase	UVW Phase
Volts (V)	5000.00	5000.00	5000.00	5000.00
Cap (pF)	668000.00	668000.00	668000.00	1994000.00
IR	1321.00	1418.00	1154.00	857.00
PI	5.03	5.18	4.78	4.64
T1	17.47	12.27	14.66	15.26
T2	103.22	90.31	93.36	104.20
T3	543.60	513.77	541.71	776.96
Q1	7.78	6.81	8.28	6.40
Q2	7.79	8.72	10.12	4.11
Q3	12.15	12.18	10.78	6.71
Q1/Q2	1.00	0.78	0.82	1.56
Q2/Q3	0.64	0.72	0.94	0.61
AF	62.63	47.01	50.95	60.32
CI	0.03	0.02	0.02	0.02
IC/IR	-2.61	-4.28	1.91	-2.26
DF	1.28	1.28	1.29	1.17
I	0.42	0.38	0.62	0.61
ROVOL	2.54e+14	2.83e+14	1.73e+14	5.25e+14
ROVOL (log10)	14.40	14.45	14.24	14.72
Q1T1/Q2T2	0.17	0.11	0.13	0.23

PDCA Test Results - U Phase

PDCA Test Results - V Phase

PDCA Test Results - W Phase

PDCA Test Results - UVW Phase